# Multimodal-Input-Driven Editable and Controllable 3D Object Manipulation

Chia Hui Yen, David Chen, Karthick Raja BG, Yizhuo Di
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## 1 Project Goal and Motivation

The goal of this project is to build a model that enables text-driven editing of existing 3D CAD models, specifically focusing on parametric B-rep (Boundary Representation) structures. Given an input CAD model and a textual instruction describing the desired modification, the system should generate a new B-rep model that reflects the edit while maintaining topological validity and preserving the original structure where appropriate. This approach aims to make CAD editing intuitive and semantically meaningful, reducing the need for manual parametric manipulation and bridging the gap between natural language expression and geometric transformation. As a fallback, we will also explore text-driven mesh editing for unstructured 3D representations when CAD-specific approaches face integration or data challenges.

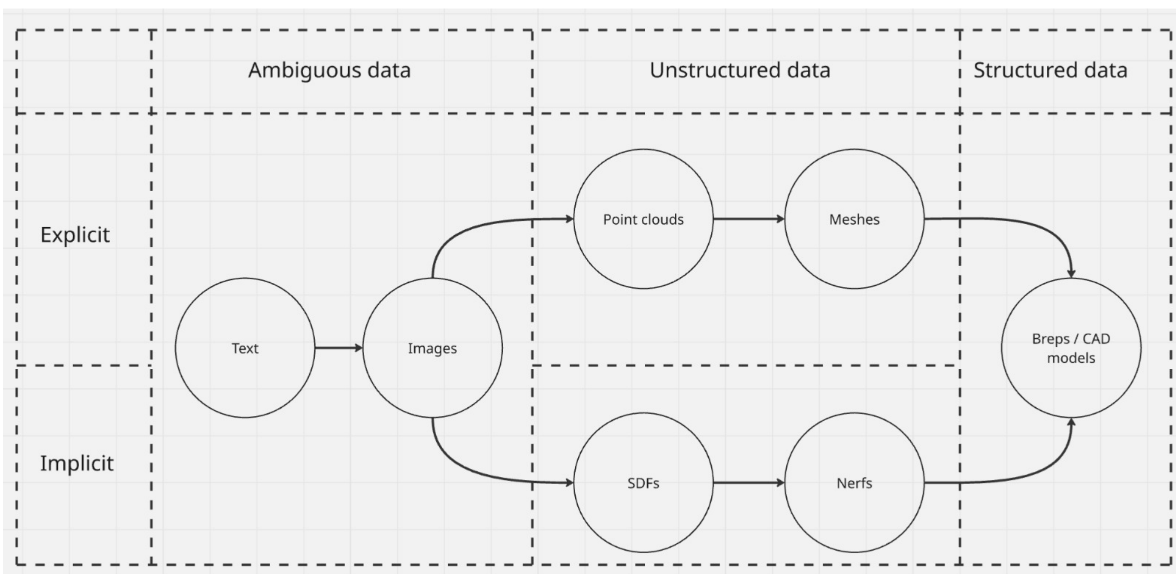


Figure 1: Overview of modality spectrum from ambiguous (text, images) through unstructured (point clouds, meshes, SDFs, NeRFs) to structured data (B-reps/CAD models). Our primary focus is on structured CAD models with fallback to mesh-based editing.

## 2 Plan of Work

Our primary approach targets parametric CAD models with B-rep structure. The project will develop a hybrid system where a perception model (Vision-Language Model or GNN) analyzes multimodal inputs (text prompts, spatial indicators, reference images) and generates a high-level symbolic design plan. This plan is then executed by a symbolic geometric solver that enforces mathematical precision, geometric constraints, and topological validity—addressing the precision gap and topological invalidity issues inherent in purely neural approaches.

As illustrated in Figure 1, our system will handle inputs across the modality spectrum, from ambiguous text and images to explicit structured representations. The neural planner will translate ambiguous high-level intent into concrete geometric operations, while the symbolic solver ensures the resulting B-rep satisfies CAD validity constraints. This approach draws inspiration from RLCAD [16], which demonstrates the potential of reinforcement learning for generating CAD command sequences in parametric design paradigms.

**Fallback Strategy:** If integration complexity or B-rep dataset limitations impede progress, we will pivot to a mesh-based latent editing approach. We will represent meshes in a latent space using a pre-trained 3D generative backbone such as DeepSDF or a diffusion-based model. A language encoder (CLIP or transformer-based) embeds textual instructions into the same latent space. The core model learns to predict latent transformations conditioned on text, optimizing for both geometric reconstruction and semantic alignment.

# 3 Relation to Existing Research

Recent studies have explored text-guided 3D editing by aligning natural language with 3D generative representations. FocalDreamer introduces a focal-fusion mechanism for localized shape editing from text prompts, demonstrating how language can guide precise modifications without regenerating the entire object [19]. SHAP-EDITOR presents an instruction-conditioned latent editor that performs feed-forward 3D modifications within seconds, avoiding the heavy optimization typical of earlier text-to-3D pipelines [4]. More recently, 3D-LATTE integrates textual conditioning into a 3D diffusion latent space, achieving efficient and semantically aligned editing without retraining or test-time optimization [8]. Together, these approaches reveal the potential of coupling language and geometry, though most operate on implicit neural fields rather than explicit parametric representations.

Parallel efforts have focused on structure-preserving shape manipulation in latent space. Fine-Grained 3D Vehicle Shape Manipulation via Latent Directions demonstrates interpretable geometric control by learning attribute-specific latent vectors in DeepSDF space [9], while VoxHammer achieves precise and coherent 3D edits through geometry-aware denoising in voxel space [13]. However, these methods lack textual conditioning and remain domain-specific.

For parametric CAD modeling, RLCAD [16] demonstrates a reinforcement learning approach to generate CAD command sequences, showing that a typical CAD command sequence represents a parametric design paradigm. While RLCAD focuses on generating complete designs from scratch, our project aims to extend these ideas toward text-driven editing of existing CAD models, where an input model and natural-language instruction jointly drive transformations that preserve validity and structural identity.

## 3.1 Scope Differentiation

Our scope differs in category, modality, control, and evaluation. We primarily target structured data (CAD models, B-rep) with unstructured data (meshes, point clouds) as backup. We introduce auxiliary modalities to constrain and edit precisely: text prompts for high-level intent, spatial indicators to localize and bound edits (bounding boxes, segmentation masks) for targeted editing when needed [2, 19], and data indicators for specific edit parameters [12]. Our domain focus is on CAD models in industrial/architectural contexts where B-reps are standard, prioritizing datasets with part/feature semantics to support precise editing and evaluation.

# 4 Novel Approach and Key Ideas

## 4.1 Gaps and Limitations We Target

Current methods are insufficient because they struggle with several fundamental challenges that our CAD-focused approach aims to address directly:

**Topological Invalidity:** Direct generative models often produce invalid geometry (e.g., gaps, non-manifold edges) that is unusable in engineering workflows. CAD models require watertight, manifold surfaces.

**The Precision Gap:** Neural networks are inherently approximate, failing to meet the micron-level precision and mathematical constraints (e.g., perfect tangency, parallel surfaces) required by CAD. Our symbolic solver addresses this limitation.

**The Complexity Ceiling:** Existing models are largely confined to simple sketch-and-extrude operations, unable to generate parts using more complex commands like lofts, sweeps, or revolutions due to data limitations and lack of parametric understanding.

Our approach addresses these gaps by targeting structured CAD/B-rep models with a hybrid neural-symbolic system. We introduce auxiliary modalities to reduce ambiguity: text prompts provide high-level intent, spatial indicators (bounding boxes, segmentation masks) localize edits, and data indicators specify edit parameters. The goal is to prevent unintended deformations while maintaining CAD validity.

## 4.2 Core Insights and Constraints

The major constraints are fidelity, controllability, and topological validity—particularly critical for CAD applications. Our core insight is to create a hybrid system that leverages the "best of both worlds." Instead of a single end-to-end model, our approach decouples the problem:

**Neural Perception:** A perception model (Vision-Language Model or GNN) analyzes the input (point cloud, sketch, or text) and generates a high-level, symbolic design plan (e.g., "a cylinder joined to a cube with a filleted edge").

**Symbolic Execution:** This plan is executed by a symbolic geometric solver—a deterministic component responsible for enforcing mathematical precision, geometric constraints, and ensuring the final B-Rep is topologically valid. This approach is inspired by RLCAD's demonstration that CAD command sequences can effectively represent parametric design paradigms [16].

This division of labor allows the neural network to focus on creative inference and pattern recognition, while offloading the requirements of precision and validity to a system built for logical rigor. This is particularly crucial for CAD applications where approximate solutions are unacceptable.

## 4.3 Risks and Fallback Plans

**Risk 1: Integration Complexity**

The primary risk is the integration complexity between neural planner and symbolic solver. Creating a seamless and robust interface is a significant engineering challenge. The symbolic solver might frequently fail if the neural network's high-level plans are ambiguous or geometrically impossible to resolve.

*Fallback:* If integration proves too complex, we pivot to the mesh-based latent editing approach described in Section 2, sacrificing CAD precision for demonstrable semantic editing capabilities.

**Risk 2: Multimodal Dataset Availability**

One of the most significant challenges lies in assembling or curating a multimodal dataset that includes aligned text, geometry, and semantic labels. Existing datasets primarily consist of unstructured 3D representations (e.g., point clouds, meshes) rather than parametric CAD/B-rep models, limiting geometric precision and editability. Datasets rarely contain text + geometry + semantic label triplets, which are crucial for training cross-modal editing or generation models. Reconstructed CAD/B-rep models are seldom openly available, making data standardization and interoperability difficult.

*Fallback:* Following the strategy in Miao et al. (2024) [10], use a 3D diffusion model for generation, but adopt Miao-style attribute regression in latent space to steer edits. Concretely, train an attribute regressor on the diffusion latent (or intermediate features) and use it to (a) define attribute directions for latent updates or (b) provide attribute-guided conditioning (e.g., via classifier(-free) guidance) during sampling. This ports the 'regression-guided latent editing' idea to a diffusion backbone, operating on meshes rather than B-reps.

**Risk 3: Editing and Control Risks – Spatial Localization and User Control**

Multimodal editing accuracy depends heavily on the model's ability to localize editing regions and interpret ambiguous language inputs. Text prompts such as "extend the top surface slightly" lack geometric specificity, often resulting in over-editing or unintended deformations. Over-reliance on textual cues without geometric anchors also risks poor spatial alignment.

*Fallback:* Adopt a bounding-box or segmentation-guided editing workflow inspired by FocalDreamer (Zhang et al., 2023) [19] and SceneFactor (Bokhovkin et al., 2024) [2]. These methods employ region masks, ellipsoid focal regions, or scene factors to confine local edits, ensuring geometric consistency and part-wise control even when text descriptions are vague. This fallback allows the system to leverage 2D images or mask inputs as auxiliary modalities until full text-to-3D alignment is achieved.

# 5 Evaluation Methodology, Datasets, and Metrics

## 5.1 Training Objectives

The training framework will combine geometric reconstruction with semantic control mechanisms. For the CAD-focused approach, we will develop supervision signals for symbolic plan generation. For the mesh-based fallback, we anticipate the following loss components:

**Geometric Reconstruction Loss:** Chamfer Distance (CD): $L_{CD} = \sum_{x \in S_1} \min_{y \in S_2} ||x-y||^2 + \sum_{y \in S_2} \min_{x \in S_1} ||y-x||^2$. Earth Mover's Distance (EMD) as an alternative for global shape fidelity. Reference: Fan et al. [5].

**Attribute Mapping Loss:** Supervised regression between latent codes and semantic attributes: $L_{attr} = ||f_\theta(z) - a||^2$ where $f_\theta$ is a learned regressor, $z$ is latent code, $a$ is target attribute. Reference: Miao et al. [10].

**Disentanglement Regularization:** Orthogonality constraint between attribute directions: $L_{ortho} = \sum_{i \neq j} |\langle d_i, d_j \rangle|$ where $d_i$ represents learned direction vectors. Alternative approaches may include $\beta$-VAE style regularization or contrastive learning. References: Higgins et al. [6]; Chen et al. [3].

**Edit Locality Loss:** Spatial masking to constrain edits: $L_{local} = ||M \odot (S_{edit} - S_{orig})||^2$ where $M$ is a binary spatial mask.

**Cross-Modal Alignment:** CLIP-based text-geometry alignment: $L_{CLIP} = 1 - \cos(E_{text}(p), E_{img}(R(S)))$ where $R$ renders the shape, $E$ are CLIP encoders. Reference: Radford et al. [11].

## 5.2 Quantitative Evaluation Metrics

**Set-Level Generative Metrics:** Maximum Mean Discrepancy (MMD), Coverage (COV), 1-Nearest Neighbor Accuracy (1-NN) test for mode collapse and overfitting. References: Achlioptas et al. [1]; Yang et al. [17].

**Edit-Specific Metrics:** Edit Locality (percentage of geometric change in target region), Attribute Alignment (Spearman correlation between control parameters and measured attributes), Identity Preservation (Chamfer Distance outside edit region).

**CAD Validity Metrics:** Topology Validity (watertight surface check, manifold verification), Novelty/Uniqueness as defined in Willis et al. [14] and Xu et al. [15]. These metrics are critical for the CAD-focused approach.

## 5.3 Qualitative Evaluation

**Visual Assessment:** Side-by-side comparisons of original and edited geometry, multi-view renderings for 3D consistency, interpolation sequences between attribute extremes.

**Ablation Studies:** Effect of disentanglement regularizers on attribute independence, impact of locality constraints on edit precision, comparison of different attribute parameterizations. For CAD approach: effectiveness of symbolic solver vs. pure neural methods.

**Comparative Baselines:** Naive latent space arithmetic, Miao et al.'s vehicle editor (if replicable) [9], text-to-3D editing methods (e.g., FocalDreamer) [19], and RLCAD's parametric generation approach [16].

**User Study (time permitting):** Task-based evaluation with controllability ratings (Likert scale), task completion time, preference rankings. Sample size: target 15-20 participants.

## 5.4 Datasets

**Primary (CAD-focused):**

**ABC Dataset** (`https://deep-geometry.github.io/abc-dataset`): 1M+ CAD models with B-Rep structure. Essential for parametric/CAD editing. License: MIT.

**Fusion 360 Gallery** (`https://github.com/AutodeskAILab/Fusion360GalleryDataset`): 8,625 parametric CAD assemblies with reconstruction sequences. High-quality engineering models. License: Non-commercial research.

**Fallback (Mesh-based):**

**ShapeNet Core** (`https://shapenet.org`): 51,300 3D models across 55 categories. Focus categories: cars (7,497), chairs (6,778). License: Research/non-commercial use.

**PartNet** (`https://partnet.cs.stanford.edu`): 26,671 3D models with fine-grained part-level annotations. Essential for part-based editing and locality evaluation. License: MIT.

**Objaverse** (`https://objaverse.allenai.org`): 800K+ diverse 3D assets (potential scale-up dataset). Would require quality filtering and subset selection.

We anticipate primary reliance on ABC Dataset + Fusion 360 Gallery for CAD approach, with fallback to ShapeNet + PartNet for mesh-based editing.

## 5.5 Success Criteria

**Minimum:** Demonstrate controllable attribute editing on CAD models or a single mesh category (cars/chairs) with quantitative improvement over naive latent interpolation in at least 2 of 3 metrics (MMD, edit locality, attribute alignment). For CAD: demonstrate topologically valid outputs.

**Target:** Achieve text-driven CAD editing with high topology preservation rate (¿85%) and precision within engineering tolerances. If pursuing mesh fallback: achieve comparable or better controllability than Miao et al. baseline while improving disentanglement.

**Stretch:** Enable cross-modal control (text + spatial indicators + sketches) with good CLIP alignment scores for CAD models, demonstrating seamless integration of neural planner and symbolic solver.

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